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Nontraditional Risk Factors for Peripheral Arterial Disease: The Evidence and Underlying Mechanisms.

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Role of Depression and Anxiety in Peripheral Artery Disease: Correlates, Outcomes, and Considerations for Treatment

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ABSTRACT: Peripheral artery disease (PAD) affects millions worldwide and is associated with high mortality, major amputations, and detrimental effects on functional and health status. PAD management is multifactorial and encompasses addressing cardiovascular and metabolic risk factors, improving symptoms and vessel patency through exercise therapy or revascularization, and medications. PAD frequently develops under complex biopsychosocial circumstances, and those same mechanisms impact treatment and outcomes. In these contexts, depression and anxiety play a significant role and are known to be associated with a higher risk of developing PAD and worse disease and treatment outcomes. Incorporating depression and anxiety assessment and treatment can complement the traditional approach to cardiovascular disease management to improve outcomes, given that we know they influence the development and progression of PAD and its treatment. In this review, we evaluate various biological and behavioral mechanisms linking depression, anxiety, and PAD, as well as the role of depression and anxiety in treatment planning. We stress the importance of considering these disorders when managing PAD with both medical and surgical approaches and discuss future avenues for a holistic, team-based, and multidisciplinary approach to improve long-term outcomes in individuals with PAD.

Key Words: anxiety ■ depression ■ health status ■ mortality ■ peripheral arterial disease

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Peripheral artery disease (PAD) is a chronic atherothrombotic occlusive disease of the arteries supplying blood to the legs, estimated to affect ≈220 million individuals globally.[1] PAD causes a wide array of symptoms ranging from asymptomatic presentation to walking-induced leg pain, ischemic rest pain, ulcers, and gangrene. The severity of walking impairment is closely linked to a reduction in patients' health status.[2] The natural history and complications of the disease are burdensome and devastating for patients, and may include recurrent infections, poor wound healing, major amputations, and progressively declining functional status.[3,4]

The high burden of symptoms, toll of associated comorbidities, and the impact on health status partially explain the considerable mental health burden associated with PAD.[5,6] However, individuals with preexisting depression or other psychiatric conditions also have a markedly increased risk of developing atherosclerotic cardiovascular disease. Accumulating evidence has demonstrated shared genetic loci between depression and anxiety and the etiological factors of the atherothrombotic, inflammatory, and endothelial degenerative processes underlying atherosclerotic disease and PAD.[5,7,8] As such, the prevalence of mental health burden in patients with PAD is higher than in the age-matched

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Nonstandard Abbreviations and Acronyms

ASIC	acid-sensing ion channel
ASIC3	acid-sensing ion channel subtype 3
GLP-1	glucagon-like peptide 1
NGF	nerve growth factor
PAD	peripheral artery disease
PAR2	proteinase-activated receptor 2
RCT	randomized controlled trial
TNFR1	tumor necrosis factor receptor 1
TNF-α	tumor necrosis factor alpha
TRP	transient receptor potential
TRPA1	transient receptor potential ankyrin 1
TRPV1	transient receptor potential vanilloid 1

overall population.[9] Importantly, the incidence and progression of PAD as well as the mental health impacts are heavily influenced by behavioral factors, including patient adherence to treatment regimens and the ability to engage in cardiovascular risk factor management, including lifestyle modification.

Although endovascular and open surgical revascularization are the mainstay invasive treatment modalities for management of chronic limb-threatening ischemia, the medical management of PAD is largely focused on lifestyle and pharmacological interventions that seek to control underlying modifiable cardiometabolic risk factors.[13,14] Center-based supervised exercise and home exercise programs have been shown to improve walking distance and health status; however, barriers to accessing such programs, as well as poor adherence, result in underutilization.[14–16] In patients who experience mental health disorders, the ability to adhere to treatment regimens that foster positive health behavior change is a unique challenge. Current care models are evolving towards establishing a multidisciplinary approach to PAD care that integrates medical management, rehabilitation, self-management strategies, and mental health treatment necessary to improve outcomes for patients with comorbid mental illness.

In this review, we aim to explore the relationship between PAD and mental illness, with a focus on depression and anxiety. Our objectives are to (1) review the foundational biological mechanisms that underlie the association between depression, anxiety, and PAD at the cardiovascular system, cellular, and tissue level; (2) examine the behavioral factors that contribute to worsened clinical outcomes in patients with PAD and comorbid mental illness; and (3) discuss how depression and anxiety impact the current management of PAD. In doing so, we highlight the case for an integrated care approach that has systems in place to address anxiety and depression in the context of PAD patient care needs.

Highlights

- Depression and anxiety are common in peripheral artery disease and independently predict worse limb and survival outcomes.
- Autonomic, inflammatory, and behavioral pathways provide mechanistic links between mental health and peripheral artery disease progression and treatment failure.
- Routine screening and integrated behavioral care are actionable strategies that may lead to improved health status and clinical outcomes in peripheral artery disease.

BIOLOGICAL MECHANISMS

Numerous biological mechanisms contribute to the association between psychological stress and atherosclerosis.[10] Chronic stress, including stress due to affective disorders and other psychiatric conditions, leads to activation of the hypothalamic-pituitary-adrenal axis and the sympathetic nervous system, which may result in downstream sleep disturbances, reduced physical activity, changes in appetite and diet, and substance use.[11–13] Prolonged activation and the release of stress hormones such as adrenaline and cortisol place the system in a persistent fight or flight mode. This imbalance results in heightened physiological responsiveness to acute stress, which may result in insulin resistance, obesity, hypertension, increased inflammation, platelet reactivity, and endothelial dysfunction.[14–16] Autonomic dysregulation has also been noted in patients with PAD and is frequently evaluated with an abnormal heart rate variability.[17–19]

Numerous studies have utilized various animal models to investigate the pathophysiological mechanisms underlying PAD. These models typically involve arterial occlusion techniques to simulate ischemic conditions and study their effects on sympathetic nerve activity, vascular responses, and exercise intolerance. Here, we review the main mediators studied in the regulation of blood flow by the autonomic nervous system: transient receptor potential (TRP) channels; acid-sensing ion channels (ASICs); dysregulation of neurotransmitters and adrenergic receptors; and regulation of collateral blood flow. These players may be dysregulated by psychological stress and can explain the biological link between mental illness and the progression of symptoms in PAD.

The TRP channels, including TRPV1 (TRP vanilloid 1) and TRPA1 (TRP ankyrin 1), are cation channels that are involved in cellular signaling to facilitate several sensory and physiological processes. TRPV1 is sensitive to temperature and pain, whereas TRPA1 responds to mechanical stimuli and chemosensation. Both channels are located in sensory neurons and have a major role in the autonomic nervous system and blood flow.[20] These

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channels are integral to sensory neurons that innervate muscles and modulate sympathetic activity after peripheral arterial occlusion. Upon arterial occlusion, TRPV1 expression in sensory neurons is upregulated, leading to increased sympathetic reflex responses.[21] TRPA1 channels on sensory muscle afferent neurons similarly contribute to amplified sympathetic responses during exercise. The interaction between TRPA1 and PAR2 (proteinase-activated receptor 2) regulates this abnormal function, with enhanced PAR2 expression in the dorsal root ganglia of the occluded limb increasing TRPA1 currents and the pressor response during exercise.[22] TRPA1 upregulation in dorsal root ganglia sensory neurons after femoral artery occlusion leads to increased sympathetic activity and blood pressure responses, particularly during static muscle contraction.[23] TRP channels have also been implicated in anxiety and depression, and genetic deletions of these channels differentially impact susceptibility to these disorders in mice.[24]

ASICs are a type of ion channel that is activated by acidic conditions. These channels are expressed throughout the central and peripheral nervous system. They play a significant role in detecting changes in pH and are activated in various physiological and pathological processes, such as pain, learning, memory, and fear.[25] ASIC antagonism with amiloride significantly attenuates pressor and sympathetic responses to skeletal muscle metabolic reflex activation in humans.[26] This finding supports the role of ASICs in mediating sympathetic nerve activity and suggests that targeting these channels could be beneficial in managing exaggerated autonomic responses in PAD.

Inflammatory biomarkers and the interplay between neurotrophic and inflammatory factors, notably NGF (nerve growth factor) and TNF-α (tumor necrosis factor alpha), modulate sympathetic nervous system activity in PAD. The involvement of NGF in enhancing sympathetic and blood pressure responses, particularly through its effect on ASIC3 (ASIC subtype 3), underscores its potential role in PAD.[27] Concurrently, the elevated levels of TNF-α and its receptor, TNFR1 (tumor necrosis factor receptor 1), in the dorsal root ganglion of ischemic limbs point to an inflammatory mechanism exacerbating sympathetic activity, further supported by the attenuation of these responses on inhibition of TNF-α synthesis.[28]

The mechanistic roles of neurotransmitters and their receptors, particularly norepinephrine and α-adrenergic receptors, in PAD are demonstrated through their impact on vascular resistance and sympathetic nerve function under ischemic conditions. Acute ischemic conditions result in decreased skeletal muscle norepinephrine levels, indicative of impaired neurotransmitter release and axonal conduction.[29,30] This reduction in norepinephrine release suggests an immediate ischemic impact on nerve function, leading to diminished sympathetic outflow. As ischemia progresses, an observed increase in norepinephrine levels suggests a shift toward compensatory mechanisms, likely mediated by intracellular calcium overload, transitioning from impaired axonal conduction to enhanced norepinephrine exocytosis.[29,30] The α-adrenergic receptors also play a pivotal role in modulating vascular responses to ischemia. The activation of α1- and α2-receptors has been shown to differentially influence vascular resistance and blood flow in ischemic conditions. α1-Receptor inhibition increases blood flow to muscles unaffected by ischemia, whereas α2-receptor inhibition influences both collateral and muscle conductance, indicating a complex regulatory role of these receptors in maintaining vascular tone during PAD.[31]

Recent studies have also demonstrated that sympathetic stimulation leads to constriction of collateral vessels, reducing their conductance and consequently the efficiency of blood flow to the limb. This effect is more pronounced in atherosclerotic models, underscoring the impact of sympathetic nerve activity on vascular function in PAD.[32] The role of the sympathetic nervous system in peripheral circulation is exemplified in patients with advanced PAD who undergo sympathectomy. Case series have shown that sympathectomy provides relief from rest pain and increases skin blood flow and warmth in the lower extremities, with over 70% of patients experiencing symptomatic improvement.[33]

An additional body of evidence relates the exposure to trauma and posttraumatic stress as an independent risk factor in developing cardiovascular disease.[34–36] Pathophysiologic mechanisms underlying this relationship that have emerged in the literature include promotion of oxidative stress and inflammation, hormonal dysregulation, epigenetic changes, and resultant vascular remodeling.[37,38]

Collectively, these described mechanisms involving exaggerated autonomic nervous system responses may partially overlap with the stress model for atherosclerosis, and may indirectly also explain the link between anxiety and depression and adverse PAD outcomes, although much mechanistic evidence specific to PAD and the link with anxiety and depression is missing and is derived from other cardiovascular diseases.[11–13]

BEHAVIORAL MECHANISMS RELATED TO ANXIETY AND DEPRESSION RELEVANT TO PAD OUTCOMES

Clinically relevant symptoms of anxiety and depression may reinforce the expression of established behavioral risk factors for cardiovascular disease, such as poor diet, physical inactivity, poor adherence, and substance use disorders. For example, a recent study suggested that a decrease in physical activity precedes an increase in depressive symptoms over a 1-year period.[39] Furthermore, pain syndromes in PAD are highly prevalent and

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further contribute to increased substance use. High sustained opioid use is seen in at least 1 quarter of individuals with PAD, and this risk is further exacerbated by revascularization interventions.[40] The compounding nature of these comorbidities contributes to worsened clinical outcomes, as patients with comorbid depression and opioid use disorder face increased length of stay and increased risk of hospital readmission after vascular intervention.[41] The complex interplay between comorbid depression, anxiety, pain, and substance use thus creates a distinct impact on a patient's overall clinical trajectory and outlook.

Depression and anxiety are among the most predictive factors for poor health status in patients with PAD.[42] Instruments to assess depression and anxiety in clinical research include, for example, the Patient Health Questionnaire for depressive symptoms and the General Anxiety Disorder questionnaire for anxiety symptoms, which are among the most widely used instruments to screen for depression and anxiety, respectively.[43–47] In a multicentric multinational PAD cohort of patients being followed in specialized vascular centers, the Patient Health Questionnaire-8 and Generalized Anxiety Disorder-2 scores were the second and third most predictive factors of PAD-specific health status at 1 year.[42] Moreover, patients with documented depression have worsened trajectories in health status over 1 year,[48] but patients who were depressed and experienced an improvement in their depression based on their Patient Health Questionnaire 8 scores relative to their baseline score also experienced health status gains at 1 year after angioplasty compared with those who continue to be depressed.[49] Anxiety is not only a predictor of health status but has also been shown that despite revascularization and management of PAD with subsequent improvement of generic health status, anxiety symptoms remain high.[50]

INTERACTIONS WITH PAD MANAGEMENT

Periprocedural and long-term anxiety have been linked to multiple in-hospital interactions after revascularization. Anxiety has been associated with increased length of stay and healthcare costs among patients undergoing endovascular revascularization.[51] More recent trials have also focused on the periprocedural anxiety to improve short and long-term revascularization outcomes. A recent small randomized controlled trial demonstrated that patients who listened to classical music in the preprocedural space before a peripheral vascular intervention recovered from anxiety more quickly than controls.[52] Early studies have proposed the use of mind-body interventions incorporating elements of mindfulness and guided imagery to target periprocedural anxiety, interoceptive awareness, quality of life, and intervention acceptability.[53] These interventions have the potential to not only improve periprocedural anxiety, and the degree of sedation requirements, but also may aid in the improvement of long-term outcomes after revascularization.

Outcomes after revascularization may also differ among patients with depression (Table). Depressive symptoms correlate with the number of prior vascular procedures and amputations, as well as increased pain levels at admission.[54] In-hospital outcomes after revascularization (either endovascular or surgical) are also worse among patients with depression. Specifically, a nationwide study found that depression was associated with an increased odds of major amputation and increased length of stay regardless of the revascularization strategy.[51] Furthermore, patients with depression are 1.5 times more likely to be discharged to a nursing or acute care facility after revascularization.[55] The negative impacts of depression on PAD outcomes seem to be attenuated by treatment with antidepressants. For example, in 1 cohort study, the risk of amputation in patients with PAD and untreated comorbid depression was higher than in patients on antidepressant therapy (42% versus 10% higher than those without depression).[56] In addition to the severe toll on disease morbidity, depression is associated with a 24% increased risk of all-cause mortality in patients with PAD.[57]

Depression and anxiety may also promote negative illness perceptions surrounding PAD,[58] thus creating specific barriers to medication adherence, appointment follow-up, and engagement in rehabilitation programs such as supervised exercise regimens. Depression has been independently linked to lower adherence to medication use in chronic diseases,[59] further undermining chances for optimal adherence to guideline-directed medical therapy.[60,61] Practice guidelines for PAD additionally emphasize the need for strict control of lifestyle comorbidities such as smoking cessation, management of hypertension and diabetes, and preventative foot care, all of which require patient participation and active self-management.[62] Low rates of self-management and perceptions of poor self-efficacy are seen in patients with mood disorders who suffer from other chronic illnesses, thus compromising patients' ability to achieve behavior change.[63]

INTEGRATED CARE PATHWAYS FOR DEPRESSION AND ANXIETY

As the case for concurrent behavioral interventions for depression and anxiety alongside treatment for PAD can be made from multiple angles, appropriate care pathways need to be designed to address these comorbidities. Recent calls to action from the vascular community have proposed integrating psychosocial care with the goals of (1) addressing comorbid mental health conditions, (2) promoting healthy behaviors; (3) evaluating the physical, emotional, and behavioral aspects of the pain experience; and (4) recognizing the broader impact of

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Table. Selected Studies and Reviews Assessing the Role of Depression on Outcomes in PAD

Authors	No. of studies	Size of study population (n)	Year range	Prevalence of depression	Setting	Outcomes	Study type
Scierka et al[57]	4	119 123	2000–2014	16.2%	United States, the Netherlands	Increased risk of all-cause mortality (HR, 1.24 [95% CI, 1.07–1.25]; *P*=0.005)	Systematic review and meta-analysis
						No difference in MALE	
Abi-Jaoudé et al (2022)	7	338 696	1998–2014	13%	United States	20% increase in MALE (RR 1.20 [95% CI, 1.11–1.31]; *P*<0.001)	Systematic review and meta-analysis
						No mortality difference	
						No difference in MACE	
Grenon el al (2012)	1	1024	2000–2002* recruitment	19%	United States	Increased PAD events (HR, 2.09 [95% CI, 1.09–4.00]; *P*=0.03)	Secondary analysis
McDermott et al (2003)	1	423	1998–2000	21.7%	United States	Shorter 6-minute walk distance (*P*<0.001), slower usual-pace walking velocity (*P*=0.005), slower fast-paced walking velocity (*P*=0.005)	Cross-sectional study
Harris et al[6]	1	1 415 888	2011–2017	12–14%	United States	Higher odds of amputation after both surgical and endovascular revascularization	Abstract
						Longer LOS (*P*<0.0001)	
						Higher cost of stay (*P*<0.0001)	
Zahner et al (2020)	1	116 008	2012–2013	9.1%	United States	Higher odds of major amputation (adjusted OR, 1.39 [95% CI, 1.30–1.49]; *P*<0.001)	Observational study
						Longer LOS (*B*=0.199 [95% CI 0.155–0.244]; *P*<0.001)	
Arya et al[56]	1	155 647	2003–2014	16%	United States	13% higher risk of amputation (HR, 1.13 [95% CI, 1.07–1.19])	Observational study
						17% higher risk of mortality (HR, 1.17 [95% CI, 1.14–1.20])	
Ramirez et al[55]	1	64 817	2012–2014	8.4%	United States	Increased odds of postoperative nonhome discharge (OR, 1.50 [95% CI, 1.40–1.61])	Observational study

HR indicates hazard ratio; MACE, major adverse cardiac event; MALE, major adverse limb event; OR, odds ratio; and PAD, peripheral artery disease.
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PAD, including coping mechanisms and disease adaptation.[64] Integrated depression treatment through collaborative protocols has previously been successful in improving not only depressive symptoms but also metabolic outcomes, and reducing cost and improving quality of life in primary care settings.[65–69] However, levels of integration may vary depending on resources available and institutional capabilities, ranging from colocation to full integration of the behavioral health specialist in the treatment team or via telehealth. Although integrated care services for behavioral health are billable services for licensed behavioral health professionals in the United States, reimbursement rates by insurance are relatively low, and resource redistribution at the health system level and expansion of coverage are areas that would further solidify the integration of mental health services in the PAD specialty setting.

Comprehensive evaluation of the nature, the intensity, and duration of anxious and depressive symptoms should ideally occur concurrently to the evaluation of PAD-related symptoms, depending on the resources available within the practice, and the clinical pathways for care linkage on positive screening. Additional comorbidities that may explain anxiety or depression also need to be also ruled out, such as thyroid dysfunction or nutritional deficiencies. Clinical assessment of depression and anxiety commonly incorporates screening tools such as the Patient Health Questionnaire 9 and Generalized Anxiety Disorder 7 questionnaires. In the event of a positive screen, a further clinical diagnostic interview is indicated. Evidence-based treatments for depression and anxiety in cardiovascular disease include cognitive-behavioral therapy approaches, which promote cognitive restructuring, behavioral activation, and problem-solving strategies, and exposure-based processing for anxiety.[70,71] Other therapies such as mindfulness-based approaches, acceptance and commitment therapy, eye movement desensitization reprocessing (a type of psychotherapy that helps individuals process traumatic memories by recalling distressing events while performing guided bilateral stimulation to facilitate adaptive reprocessing), and interpersonal therapy have also been explored.[72,73] Patient preferences and severity of the conditions may necessitate pharmacological management of anxiety and depression, with selective serotonin reuptake inhibitors being the most researched and best demonstrated safety profile for cardiovascular populations.[73] Integrated care teams may need to liaise with a consultant psychiatrist as well as primary care providers to have these prescribing needs met and monitored. Moreover, emerging newer cardiovascular

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therapies such as GLP1 (glucagon-like peptide 1) agonists may ultimately benefit from a cardiovascular as well as a psychological standpoint indirectly by increasing walking distance and lower extremity perfusion noted in recent trials.[74,75] Finally, resources and protocols need to be in place to provide urgent psychiatric care in high-risk scenarios, such as in individuals expressing suicidal ideation or intent.

FUTURE DIRECTIONS

The evidence toward the impact of depression and anxiety on the incidence of PAD, their impact on PAD treatment success, and outcomes has been steadily growing. As the national landscape in cardiovascular care progresses more toward integrated approaches and cardiovascular team care,[76] integration of behavioral care needs for common clinically relevant scenarios of depression and anxiety in PAD lends itself well to offering treatment pathways that are aimed at improving wellbeing and PAD outcomes at the same time.[64,77] In addition to common mental health conditions like depression and anxiety, there is also an argument to make that PAD can be considered in many patients a chronic pain syndrome,[78] which would equally benefit from multimodal treatment approaches. Implementation and evaluation of clinical care pathways that allow for this integration and the study of its impact on outcomes are needed.

Although many of the insights into mechanisms have been derived from coronary artery disease and cardiovascular disease overall, PAD-specific mechanistic pathways that link the impact of chronic and acute stress on limb outcomes need to be further studied with a translational approach. PAD is a lifestyle disease, and behavioral factors and stress are known to play a crucial role in the disturbance of homeostasis for the hypothalamic-pituitary-adrenal axis and overactivation of the sympathetic nervous system, which has further negative effects on vascular health.[10,79,80] These pathways need to be further studied in PAD, as a way to identify modifiable targets to improve limb and cardiovascular outcomes (Figure).

Regardless of the mechanistic link with PAD progression, depression and anxiety deserve to be treated. Patients care about their quality of life, and awareness of the importance of mental health is growing in the general population, in the medical community, and as such, it is also expected to be

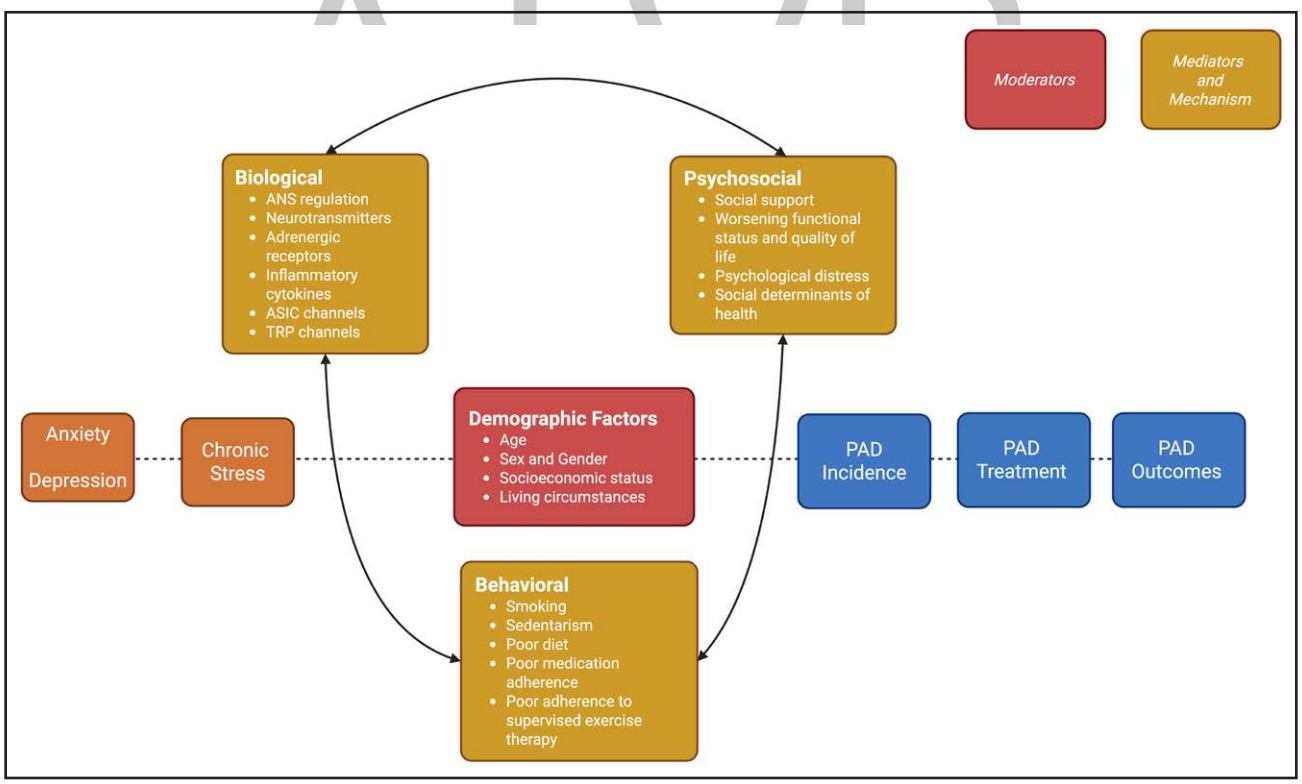

Figure. Mechanisms that mediate and moderate the link between mental health and peripheral artery disease (PAD). The presented model is based on the broader, evidence-based conceptual model developed for the association between chronic stress and atherosclerosis (Rozanski et al[80]). It states that anxiety and depression can result in exposure to chronic stress, which has downstream effects and multidirectional associations with biological factors, psychosocial factors, and impact on the behavior of individuals, through which their risk of atherosclerotic diseases, such as PAD, increases. These mechanisms have all been implicated in the increased risk of PAD incidence, PAD progression requiring treatment, and PAD outcomes. Some of these associations are moderated by demographic factors, such as age, sex, socioeconomic status, and living circumstances. As an example, individuals living in distressed neighborhoods, women, and minorities have a higher risk of both exposure to chronic stress and adverse atherosclerotic outcomes. ANS indicates autonomic nervous system; ASIC, acid-sensing ion channel; and TRP, transient receptor potential.

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addressed in the context of chronic diseases, such as PAD. The following needs are identified to realize integration of depression and anxiety risk assessment and treatment as part of the PAD clinical care trajectories: First, further advocacy and raising awareness is a shared responsibility of professional organizations, PAD care team members, which can be accomplished by, for example, offering further continuing education efforts, campaigns targeted to the public and patient communities, and embedding integrated behavioral health in PAD care as a quality or performance metric. Second, cotraining models of vascular and behavioral specialties and integration of the relationship between stress and cardiovascular disease, as part of the training curriculum of the vascular specialist, and vice versa, training behavioral specialties to understand complexities of vascular disease are important conditions to realize team-based PAD care that can offer whole person care. Third, on the research side, further mechanistic research dedicated to PAD, focusing on the intersection with mental health, would inform novel targets of treatment, and on the translational continuum, PAD-specific intervention programs that address behavioral components as part of the PAD rehabilitation trajectory would need to be developed and tested. Finally, in recognition of the case complexity and chronic care needs of this patient population, chronic disease management reimbursement codes for PAD would help in supporting building out integrated PAD care programs that can address the whole person.

ARTICLE INFORMATION

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